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🗄 SQL Data Analysis Internship– Task 1

Student Management Database

📌 **Task Description**

SQL is the language of data 🔍. In this task, you'll build a **Student Management Database** and write queries to analyze student performance.
This is your **first step into Data Analytics with SQL**, a must-have skill for any data analyst or BI professional.

🛠 Responsibilities (What to Build)

1. Database Setup

- Create database `StudentManagement`.
- Create table `Students` with fields:
  - StudentID (PK, auto increment)
  - Name, Gender, Age, Grade
  - MathScore, ScienceScore, EnglishScore

2. Insert Data

- Add at least 10 records with variety.

3. SQL Queries to Perform

- Show all student details.
- Average score in each subject.
- Top performer (highest total score).
- Count students per grade.
- Avg score by gender.
- Students with Math > 80.
- Update a student's grade.

🎯 Skills Gained

- ✅ SQL database creation
- ✅ Writing queries with SELECT, WHERE, GROUP BY
- ✅ Aggregation functions (AVG, COUNT, MAX)
- ✅ Filtering and updating data
- ✅ Thinking like a data analyst

📂 Sample Starter Code

```sql
-- Create table
CREATE TABLE Students (
  StudentID INT PRIMARY KEY AUTO_INCREMENT,
  Name VARCHAR(50),
  Gender CHAR(1),
  Age INT,
  Grade VARCHAR(2),
  MathScore INT,
  ScienceScore INT,
  EnglishScore INT
);

-- Query: Average Math Score
SELECT AVG(MathScore) AS AvgMath FROM Students;
```

📌 Resources

📺 Free Video Tutorials

- SQL Full Course – FreeCodeCamp
- SQL Basics – Programming with Mosh

📖 Reference Docs

- W3Schools SQL
- SQLBolt Interactive Lessons

🛠 Practice Platforms

- LeetCode SQL Problems
- Mode Analytics SQL Tutorial
- DB Fiddle → Run SQL in browser

✅ Deliverable

A `.sql` file containing:

- Database & table creation
- Insert statements
- Queries with results explained in a short report